

Q1:

1.7-3 Two inputs, temperature $T(t)$ and wind speed $V(t)$, produce an output, wind chill $W(t)$, according to $W(t) = 35.74 + 0.6215T(t) - 35.75\{V(t)\}^{0.16} + 0.4275T(t)\{V(t)\}^{0.16}$. The independent variable here is time, t . Answer the following questions yes or no, and provide mathematical justification for each answer.

- (a) Is this system BIBO-stable?
- (b) Is the system memoryless?
- (c) Is the system causal?
- (d) For simplicity, let the wind speed be constant, $V(t) = k_V$. Thus, $W(t) = k_1 + k_2T(t)$ for some constants k_1 and k_2 . Is this simplified system linear?
- (e) For simplicity, let the temperature be constant, $T(t) = k_T$. Thus, $W(t) = k_3 + k_4\{V(t)\}^{0.16}$ for some constants k_3 and k_4 . Is this simplified system linear?

a) Yes, the system is Bibo stable
for a system to be Bibo-stable, banded inputs must produce bounded outputs.

So,

as we are given with bound inputs, so they must produce bound outputs.

1. Constant term 35.74 (bounded)
2. $0.6215 T(t)$: $|0.6215 T(t)| \leq 0.6215 M$, (bounded)
3. $-35.75 [V(t)]^{0.16}$: Since $V(t) \geq 0$ (physical constraint)
4. $0.4275 T(t)[V(t)]^{0.16}$: Product of bounded terms is bounded.

Therefore, $|w(t)| \leq 35.74 + 0.6215 M_1 + 35.75 M_2^{0.16} + 0.427 M_1 M_2^{0.16} < \infty$

The system is BIBO-stable.

b) Yes, the system is memoryless.

The system is memoryless if output time 't' is only dependent upon input time 't' and not on the past like "t-1" or "t-5".

as,

$w(t) \rightarrow \text{depends upon} \rightarrow T(t), v(t)$

So, it's memoryless.

c) The system is causal because it only depends upon continuous time, and not on future. A system is causal if it depends only on inputs at times $T \leq t$ (present, past; not future).

Hence, the system is causal.

d) $w(t) = k_1 + k_2 T(t)$

where

$$k_1 = 35.74 - 35.75 k v^{0.16}$$

$$k_2 = 0.6215 + 0.4275 v^{0.16}$$

So,

No, it's not linear.

For linearity, additivity and homogeneity must satisfy Test:

- $w(\alpha T(t)) = k_1 + k_2 (\alpha T(t)) = k_1 + \alpha k_2 T(t)$

- $\alpha w(T(t)) = \alpha (k_1 + k_2 T(t)) = \alpha k_1 + \alpha k_2 T(t)$

$$W(\alpha I(t)) = k_1 + k_2 (\alpha I(t)) = k_1 + \alpha k_2 I(t)$$

$$\alpha W(I(t)) = \alpha (k_1 + k_2 I(t)) = \alpha k_1 + \alpha k_2 I(t)$$

Since,

$$W(\alpha I(t)) \neq \alpha W(I(t)), \text{ so it's not linear.}$$

e) $W(t) = k_3 + k_4 (V(t))^{0.16}$

applying super position:

$$W(\alpha V(t)) = k_3 + k_4 [\alpha V(t)]^{0.16} = k_3 + k_4 \alpha^{0.16} V(t)^{0.16}$$

$$\alpha W(V(t)) = \alpha (k_3 + k_4 (V(t))^{0.16}) = \alpha k_3 + \alpha k_4 V(t)^{0.16}$$

as, $W(\alpha V(t)) \neq \alpha W(V(t))$, hence it's not linear.

Q2: If the CT system is defined as $y(t) = tx(2t+1) + 5$

Determine the System classification:

(i) memoryless;

(ii) invertible;

(iii) causal;

(iv) stable;

(v) time invariant;

(vi) linear.

i) No, it's not a memoryless system, because it depends upon future values of t ; $y(t) \rightarrow x(2t+1)$

ii) $x(2t+1) = \frac{y(t)-5}{t} \rightarrow \text{let } T = 2t+1$

so,

$$x(T) = \frac{y(\frac{T-1}{2}) - 5}{(T-1)/2} \rightarrow \text{since a unique inverse exists, it's an invertible system.}$$

iii) No, it's not causal, because it depends upon time $T > t$.
as it depends upon future values of ' t '.

iv) No, the system is not stable. bounded input doesn't produce bounded outputs.

let $x(t) = A \rightarrow (\text{constant, bounded})$

$$y(t) = t \times (2t+1) + 5 = tA + 5$$

as $t \rightarrow \infty$, $|y(t)| = |tA + 5| \rightarrow \infty$

as unbounded output is generated, so it's unstable.

v) No, the system is time variant.

for time invariance, if $x(t) \rightarrow y(t)$, then $x(t-t_0) \rightarrow y(t-t_0)$

so,

$$y(t) = t \times (2t+1) + 5$$

if we shift $x(t) \rightarrow x(t-t_0)$

so,

$$y_1(t) = t \times (2(t-t_0)+1) + 5 = t \times (2t+1-2t_0) + 5$$

Time-shifted original output: $y(t-t_0) = (t-t_0) \times (2(t-t_0)+1) + 5$

since, $y_1(t) \neq y(t-t_0)$, the system is time varying.

vi) No, the system is not linear.

for linearity:

$$y(\alpha x) = \alpha y(x)$$

so,

$$y(\alpha x) = t (\alpha x (2t+1)) + 5 = \alpha t x (2t+1) + 5$$

$$\alpha y(x) = \alpha (t x (2t+1) + 5) = \alpha t x (2t+1) + 5\alpha$$

so,

as $y(\alpha x) \neq \alpha(y(x))$, it's not linear.

Q3:

Determine whether *the ideal time delay*

$$y(t) = x(t - t_0)$$

is

(i) memoryless,

(ii) invertible,

(iii) causal,

(iv) stable,

(v) time invariant, and

(vi) linear.

- i) No, it's not memory-less (unless $t_0 = 0$).
for a system to be memory-less, output T should be dependent upon $T \geq t$.
- ii) Yes, the system is invertible.
as each input creates, a different output so,
 $x(t) = y(t + t_0)$.
- iii) Yes, if ($t_0 \geq 0$)
so, the system only depends upon the past and present and no future values, so it is a causal system.
- iv) Yes,
if $x(t) \rightarrow y(t) = x(t - t_0)$, then
shifted input: $x_1(t) = x(t - T)$
output: $y_1(t) = x_1(t - t_0) = x(t - T - t_0)$
shifted original output: $y(t - T) = x(t - T - t_0)$
since
 $y_1(t) = y(t - T)$, the system is time invariant.

v1) applying super position.

$$\text{if } y_1(t) = x_1(t-t_0) \quad \text{and} \quad y_2(t) = x_2(t-t_0)$$

$$\bullet \quad (x_1 + x_2)(t-t_0) = x_1(t-t_0) + x_2(t-t_0) = y_1(t) + y_2(t)$$

$$\text{Homogeneity: } (\alpha x)(t-t_0) = \alpha x(t-t_0) = \alpha y(t)$$

- So the system is linear.

Q4:

1.7-4 Input voltage $x(t)$ applied to an inverting op-amp follower circuit produces output $y(t)$ according to

$$y(t+t_p) = \begin{cases} -V_{\text{ref}} & x(t) > V_{\text{ref}} \\ V_{\text{ref}} & x(t) < -V_{\text{ref}} \\ -x(t) & \text{otherwise} \end{cases}$$

where op-amp reference voltage V_{ref} and propagation delay t_p are both positive constants. Answer the following questions yes or no, and provide mathematical justification for each answer.

- (a) Is this system BIBO-stable?
- (b) Is the system causal?
- (c) Is the system invertible?
- (d) Is the system linear?
- (e) Is the system memoryless?
- (f) Is the system time invariant?

a) for any bounded input $|x(t)| < M$, the output satisfies $y(t+t_p) \leq \max(V_{\text{ref}}, M) < \infty$ for all t .

So, $y(t) \leq \max(V_{\text{ref}}, M) < \infty$, the system is BIBO-stable

b) A system is causal if, ' t ' depends only on input for $T < t$.

b) A system is causal if, y depends only on input for $T \leq t$.
here,

$$y(t+t_p) \text{ depends upon } x(t)$$

since $t = (t+t_p) - t_p$ and $t_p > 0$

so system depends upon past value of the input.

System is causal.

c) invertibility.

$$x(t) > V_{ref} \rightarrow y(t+t_p) = -V_{ref}$$

- many input values give same output so, input can't be uniquely recovered. System is not invertible.

d) linearity:

Due to piece wise saturation:

$$T[x_1 + x_2] \neq T[x_1] + T[x_2]$$

hence additivity fails,

System is not linear.

e) let:

$$t' = t + t_p \rightarrow y(t') \rightarrow x(t' - t_p)$$

so,

system is not memoryless as it depends upon past values.

f) apply time shift $x(t-t_0)$

$$y(t+t_p) = \begin{cases} -V_{ref} & , x(t-t_0) > V_{ref} \\ V_{ref} & , x(t-t_0) < -V_{ref} \\ -x(t-t_0) & , \end{cases}$$

- This is exact shifted version of original output
 $y(t-t_0+t_p)$
- So system is time invariant.